These equations have to be valid at all times; thus, we equate the coefficients of $\cos \omega t$ and $\sin \omega t$ on both sides of the equation to obtain

$$E_1 \cos \phi = a \cos\alpha - b \cos\theta \sin\alpha \quad \textbf{(22.30)}$$

$$E_1 \sin \phi = - b \sin\theta \sin\alpha \quad \textbf{(22.31)}$$

and

$$- E_2 \sin \phi = a \sin\alpha + b \cos \theta \cos\alpha \quad \textbf{(22.32)}$$

$$E_2 \cos \phi = + b \sin \theta \cos\alpha \quad \textbf{(22.33)}$$

If we square the four equations above and add, we would get

$$E_1^2 + E_2^2 = a^2 + b^2$$

which is to be expected because the total intensity of both the beams should be equal. Further, if we divide Eq. (22.30) by Eq. (22.33) and Eq. (22.31) by Eq. (22.32), we would obtain

$$\frac{E_1}{E_2} = \frac{a\cos\alpha - b\cos\theta\sin\alpha}{b\sin\theta\cos\alpha} = \frac{b\sin\theta\sin\alpha}{a\sin\alpha + b\cos\theta\cos\alpha} \quad \textbf{(22.34)}$$

Thus,

$$a^2 \sin\alpha\cos\alpha - b^2\cos^2\theta\sin\alpha\cos\alpha + ab$$

$$\cos\theta\left[\cos^2\alpha - \sin^2\alpha\right] = b^2 \sin^2\theta \sin\alpha\cos\alpha$$

Simple manipulations would give

$$\tan 2\alpha = \frac{2ab\cos\theta}{b^2 - a^2} \quad \textbf{(22.35)}$$

Example 22.5 We assume $b = a$ so that $2\alpha = \pi/2 \Rightarrow \alpha = \pi/4$ implying that the major (or minor) axis of the ellipse makes 45° with the y-axis [see Fig. 22.19 (b)]. Further,

$$\frac{E_1}{E_2} = \frac{1-\cos\theta}{\sin\theta} = \tan\frac{\theta}{2}$$

Thus, for $b = a$, and for

$$\theta = \frac{\pi}{3}, \quad \frac{\pi}{2}, \quad \frac{2\pi}{3}, \quad \frac{4\pi}{3}, \quad \frac{3\pi}{2} \quad \text{and} \quad \frac{5\pi}{3}$$

we will respectively have

$$\frac{E_1}{E_2} = +0.577,\ 1,\quad 1.732,\quad -1.732,\quad -1 \quad \text{and} \quad -0.577$$

which correspond to REP, RCP, REP, LEP, LCP and LEP respectively as shown in Fig. 22.19. For example, for $\theta = \pi/3$ we will have

$$E_x = a\cos(\omega t) \quad \text{and} \quad E_y = a\cos(\omega t - \pi/3)$$

As discussed in Example 22.1, we will have a right elliptically polarized wave as shown in Figs. 22.18(b) and 22.19 (b). From Eqs. (22.30) and (22.31) [or from Eqs. (22.32) and (22.33)] we will have

$$\tan\phi = -\frac{1+\cos\theta}{\sin\theta} = -\cot\frac{\theta}{2} \Rightarrow \phi = -\frac{\pi}{2} + \frac{\theta}{2}$$

where we have used the fact that $\cos\phi$ is positive and $\sin\phi$ is negative [see Eqs. (22.30)-(22.31)]. Therefore, in the rotated coordinates

$$E_{x'} = -E_1 \sin\left(\omega t - \frac{\pi}{6}\right) \quad \text{and} \quad E_{y'} = 1.733\, E_1 \cos\left(\omega t - \frac{\pi}{6}\right).$$

22.5 THE PHENOMENON OF DOUBLE REFRACTION LO 3

If an unpolarized light beam is incident normally on a calcite crystal [see Fig. 22.13 and 22.20(a)], it will split up into two linearly polarized beams. The beam which travels undeviated is known as the ordinary ray (usually abbreviated as the *o*-ray) and obeys Snell's laws of refraction. On the other hand, the second beam, which in general does not obey Snell's laws, is known as the extra-ordinary ray (usually abbreviated as the *e*-ray). The appearance of two beams is due to the phenomenon of double refraction and a crystal like calcite is usually referred to as a "double refracting" crystal. If we put a polaroid PP' behind the calcite crystal and rotate the polaroid (about NN') then for two positions of the

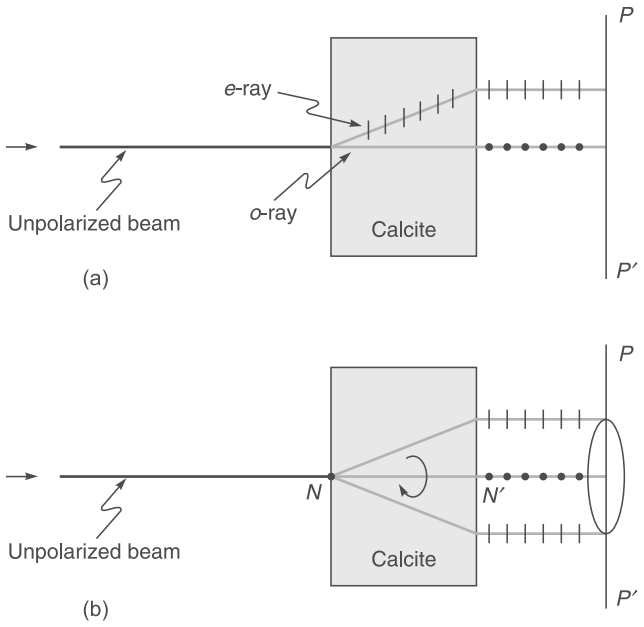


Fig. 22.20 (a) When an unpolarized light beam is incident on a calcite crystal, it usually splits up into two linearly polarized beams. (b) If we rotate the crystal about NN′ then the *e*-ray will rotate about NN′.

polaroid (when the pass-axis is perpendicular to the plane of the paper) the *e*-ray will be completely blocked and only the *o*-ray will pass through. On the other hand, when the pass-axis of the polaroid is in the plane of the paper (i.e., along the line PP') then the *o*-ray will be completely blocked and only the *e*-ray will pass through. Further, if we rotate the crystal about NN' then the *e*-ray will rotate about the axis [see Fig. 22.20 (b)]. Figure 22.21 shows a typical double image as viewed through a doubly refracting crystal like calcite. If we rotate the crystal about the vertical axis, one of the images will be fixed, while the other image will rotate.

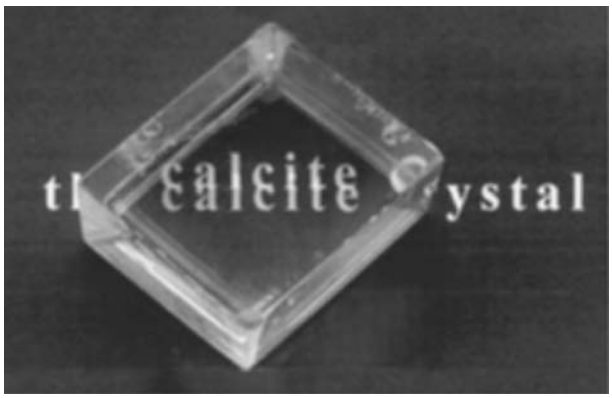


Fig. 22.21 Typical double image of a sentence in a printed text. The ordinary image is fixed, while the upper extraordinary image is shifted and can rotate. Photograph courtesy Professor Vasudevan Lakshminarayanan and adapted from Ref. 22.16; see Fig. 31 in the prelim pages.

In Sec. 22.13 we will show that whereas the velocity of the ordinary ray is the same in all directions, the velocity of the extraordinary ray is different in different directions; a medium (like calcite, quartz), which exhibits different properties in different directions, is called an anisotropic medium. Along a particular direction (fixed in the crystal), the two velocities are equal; this direction is known as the optic axis of the crystal. In a crystal like calcite, the two rays have the same speed only along one direction (which is the optic axis); such crystals are known as uniaxial crystals*. The velocities of the ordinary and the extraordinary rays are given by the following equations [see Eqs. (22.120) and (22.123)]:

$$v_{ro} = \frac{c}{n_o} \qquad \text{(ordinary ray)} \qquad \textbf{(22.36)}$$

$$\frac{1}{v_{re}^{\,2}} = \frac{\sin^2\theta}{(c/n_e)^2} + \frac{\cos^2\theta}{(c/n_o)^2} \qquad \text{(extraordinary ray)} \qquad \textbf{(22.37)}$$

where n_o and n_e are constants of the crystal and θ is the angle that the ray makes with the optic axis; we have assumed the optic axis to be parallel to the *z*-axis. Thus, c/n_o and c/n_e are the velocities of the extraordinary ray when it propagates parallel and perpendicular to the optic axis. Now, the equation of an ellipse (in the *z*-*x* plane) is given by

$$\frac{z^2}{a^2} + \frac{x^2}{b^2} = 1 \qquad \textbf{(22.38)}$$

If (ρ, θ) represent the polar coordinates, then $z = \rho\cos\theta$ and $x = \rho\sin\theta$ and the equation of the ellipse can be written in the form

$$\frac{1}{\rho^2} = \frac{\cos^2\theta}{a^2} + \frac{\sin^2\theta}{b^2} \qquad \textbf{(22.39)}$$

In three dimensions, the above equation will represent an ellipsoid of revolution with the optic axis as the axis of revolution. (If we rotate a circle about one of its diameters, we will obtain a sphere, and if we rotate an ellipse about its major (or minor) axis, we will obtain an ellipsoid of revolution). Thus, if we plot v_{re} as a function of θ, we will obtain an ellipsoid of revolution; on the other hand, since v_{ro} is independent of θ, if we plot v_{ro} (as a function of θ), we will obtain a sphere. Along the optic axis, $\theta = 0$ and

$$v_{ro} = v_{re} = \frac{c}{n_o}$$

We next consider the value of v_{re} perpendicular to the optic axis (i.e., for $\theta = \pi/2$). For a negative crystal $n_e < n_o$ and

$$v_{re}\left(\theta = \frac{\pi}{2}\right) = \frac{c}{n_e} > v_{ro} \qquad \textbf{(22.40)}$$

Thus, the minor axis will be along the optic axis and the ellipsoid of revolution will lie outside the sphere [see Fig. 22.22(a)]. On the other hand, for a positive crystal $n_e > n_o$ and

$$v_{re}\left(\theta = \frac{\pi}{2}\right) = \frac{c}{n_e} < v_{ro} \qquad \textbf{(22.41)}$$

The major axis will now be along the optic axis and the ellipsoid of revolution will lie inside the sphere [see Fig. 22.22 (b)]. The ellipsoid of revolution and the sphere are known as the ray velocity surfaces.

We next consider an unpolarized plane wave incident on a calcite crystal. The plane wave will split up into 2 plane

* In general, there may be two directions along which the two rays have the same speed; such crystals are known as biaxial crystals. The analysis of biaxial crystals is quite difficult; interested readers may look up References 22.5 and 22.6.

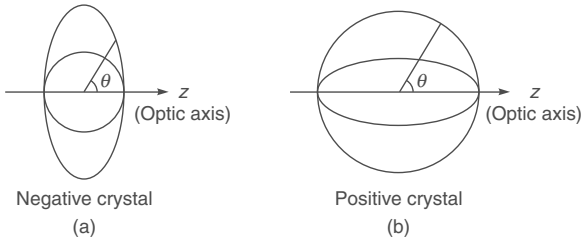


Fig. 22.22 (a) In a negative crystal, the ellipsoid of revolution (which corresponds to the extra-ordinary ray) lies outside the sphere; the sphere corresponds to the ordinary ray. (b) In a positive crystal, the ellipsoid of revolution (which corresponds to the extra-ordinary ray) lies inside the sphere.

waves. One is referred to as the ordinary wave (usually abbreviated as the *o*-wave) and the other is referred to as the extraordinary wave (usually abbreviated as the *e*-wave). For both waves, the space and time dependence of the vectors **E**, **D**, **B** and **H** can be assumed to be of the form

$$e^{i(\mathbf{k}.\mathbf{r}-\omega t)}$$

where **k** denotes the propagation vector and represents the direction normal to the phase fronts. In general, the **k** vector for the *o*- and *e*-waves will be different. In Sec. 22.12 we will show that

1. Both ordinary and extraordinary waves are linearly polarized.
2. **D . k** = 0 for both *o*- and *e*-waves **(22.42)**
   Thus, **D** is always at right angles to **k** and for this reason the direction of **D** is chosen as the direction of "vibrations".
3. If we assume the *z*-axis to be parallel to the optic axis then,
   D . ẑ = 0 (and **D . k** = 0) for the *o*-wave **(22.43)**
   Thus, for the *o*-wave, the **D** vector is at right angles to the optic axis as well as to **k**.
4. On the other hand, for the *e*-wave,
   D lies in the plane containing **k** and the optic axis (and of course, **D . k** = 0) **(22.44)**

Using the recipe given above, we will consider the refraction of a plane electromagnetic wave incident on a negative crystal like calcite; a similar analysis can be carried out for positive crystals.

22.5.1 Normal Incidence

We first assume a plane wave incident normally on a uniaxial crystal as shown in Fig. 22.23. Without loss of generality, we can always choose the optic axis to lie on the plane of the paper. The direction of the optic axis is shown as a dashed

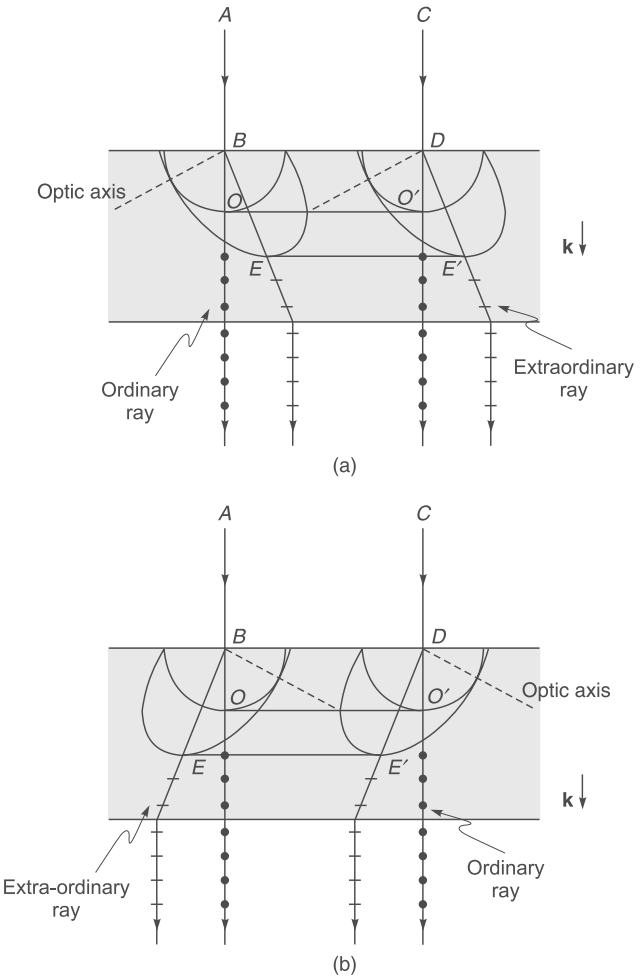


Fig. 22.23 The refraction of a plane wave incident on a negative crystal whose optic axis is along the dashed line.

line in Fig. 22.23. In order to determine the ordinary ray, with the point B as the center, we draw a sphere of radius c/n_o. Similarly, we draw another sphere (of the same radius) from the point D. The common tangent plane to these spheres is shown as OO′, which represents the wavefront corresponding to the ordinary refracted ray. It may be noted that the dots show the direction of "vibrations" (i.e., direction of **D**) which are perpendicular to **k** and to the optic axis [see Eq. 22.43].

In order to determine the extraordinary ray, we draw an ellipse (centered at the point B) with its minor axis (= c/n_o) along the optic axis and with major axis equal to c/n_e. The ellipsoid of revolution is obtained by rotating the ellipse about the optic axis. Similarly, we draw another ellipsoid of revolution from the point D. The common tangent plane to these ellipsoids (which will be perpendicular to **k**) is shown as EE′ in Fig. 22.23. If we join the point B to the point of

contact O, then corresponding to the incident ray AB, the direction of the ordinary ray will be along BO. Similarly, if we join the point B to the point of contact E (between the ellipsoid of revolution and the tangent plane EE′), then corresponding to the incident ray AB, the direction of the extraordinary ray will be along BE.

It is to be noted that the direction of **k** is the *same* for both *o*- and *e*-waves i.e., both are along BO. However, if we have a narrow beam incident as AB, then while the ordinary ray will propagate along BO, the extraordinary ray will propagate in a different direction BE, as shown in Fig. 22.23(a). Obviously, if we have a different direction of the optic axis [see Fig. 22.23(b)], then, although the direction of the ordinary ray will remain the same, the extraordinary ray will propagate in a different direction. Thus, if a ray is incident normally on a calcite crystal, and if the crystal is rotated about the normal, then the optic axis and the extraordinary ray will also rotate (about the normal) on the periphery of a cone; each time the ray will lie in the plane containing the normal and the optic axis [see Fig. 22.20 (b)].

The ray refractive index corresponding to the extra-ordinary ray (n_{re}) will be given by

$$n_{re} = \frac{c}{v_{re}} = \sqrt{n_o^2 \cos^2\theta + n_e^2 \sin^2\theta} \qquad \textbf{(22.45)}$$

If one starts with the above equation and uses Fermat's principle to obtain the refracted ray, the results will be consistent with the ones obtained in this section (see Sec. 3.5).

Now, as mentioned earlier, the direction of vibrations for the ordinary ray is normal to the optic axis and the vector **k**; as such, the directions of these vibrations in this case, will be normal to the plane of the paper and have been shown as dots in Fig. 22.23. Similarly, since the direction of vibrations for the extraordinary ray is perpendicular to **k** and lies in the plane containing the extraordinary ray and the optic axis, they are along the small straight lines drawn on the extraordinary ray in Fig. 22.23. Thus, an incident ray will split up into two rays propagating in different directions and when they leave the crystal, we will obtain two linearly polarized beams.

In the above case, we have assumed the optic axis to make an arbitrary angle α with the normal to the surface. In the special cases of $\alpha = 0$ and $\alpha = \pi/2$, the ordinary and the extraordinary rays travel along the same directions as shown in Figs. 22.24(a), (b) and (c). Figure 22.24 (b) corresponds to the case when the optic axis is normal to the plane of the paper; and as such, the section of the extraordinary

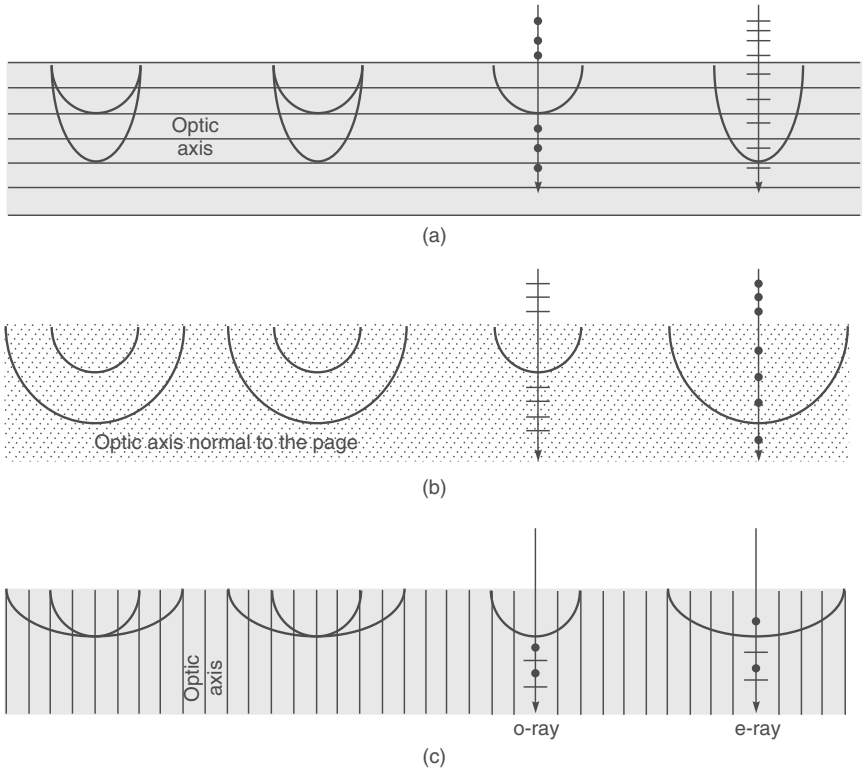


Fig. 22.24 Propagation of a plane wave incident normally on a negative uniaxial crystal. In (a) and (c) the optic axis is shown as parallel straight lines and in (b) the optic axis is perpendicular to the plane of the figure and is shown in dots. In each case, the extra-ordinary and the ordinary rays travel in the same direction.